DESIGN THINKING PROJECT



Canteen Food Ordering System

Aishwarya B
241701003

Latha A
241701024

EATALY

TAP.TASTE.GO

THE PROBLEM

Campus canteens are crowded, unpredictable, and wasteful

- Long queues during break and lunch peaks
- Students can’t see crowd levels, food availability, or prep time
- Uneven load: one canteen overloaded, another empty
- Admins have no real-time view of demand, causing food wastage

OUR PROCESS
EMPATHIZE AND DEFINE

WHO WE STUDIED ?



Students, canteen staff, admins. We interives and took survey from students of all years and staff in canteens

KEY PAIN POINTS FOUND

- “I waste half my break in the queue.”
- “I don’t know which canteen is free.”
- Staff don’t know how many orders are coming.

PROBLEM DEFINITION

The existing college canteen system is inefficient for both students and canteen administration, resulting in long waiting times, poor crowd distribution, limited order visibility, communication barriers, and difficulty managing peak-hour operations.



PIZZA SHOP
REC COLLEGE
THANDALAM CHENNAI
ITEMWISE SALE REPORT

CURRENT DAY REPORT

DT: 20/08/26 TM: 11:31:32 M.ID: 0934040266

CODE	ITEM NAME	QTY	AMOUNT
1	IDLY	77.000	770.00
5	NICE DOSAI	46.000	1840.00
8	MASALADOSAI	12.000	600.00
10	KAL DOSA	33.000	825.00
20	MEDHU VADA	40.000	600.00
24	VEN PONGAL	5.000	200.00
30	TEA	35.000	350.00
32	CREAMBUN	21.000	315.00
33	JAM+BUN	6.000	90.00
34	LEMON SEVAI	1.000	45.00
36	VEG PUFF	18.000	450.00
37	EGG PUFF	67.000	2010.00
44	CAVINS	16.000	640.00
47	DARK FANTASY	1.000	50.00
54	MUMMY MAGIC	1.000	10.00
55	HIDESEEK	5.000	50.00
56	HAPPY+HAPPY	5.000	50.00
57	MUNCH@MAX	7.000	175.00
58	KIT+KAT	2.000	40.00
59	KITKAT	2.000	100.00
60	MILKYBAR	2.000	40.00
70	LOTTE-BIG	17.000	340.00
71	SMOOTH	24.000	240.00
72	FROOTI	48.000	480.00
74	BOVONTO	16.000	480.00
75	MANGO	2.000	40.00
77	BOOST	1.000	40.00
82	TOO..YUM	23.000	460.00
85	UNIBIC	6.000	60.00
87	WATER BOTTLE	8.000	80.00
129	BREAD OMELETTE	19.000	855.00
135	BRU..COLDCOFFEE	1.000	45.00
148	SAMOSA	54.000	810.00
149	TATA	44.000	440.00
TOTAL AMOUNT :		665.000	13620.00

THANK YOU

HOW MIGHT WE...

Enable pre-ordering, convenient pickup and efficient admin management across multiple canteens?

Less waiting. Better planning.

OUR PROCESS
DESIGN PROCESS

LOGO DESIGN

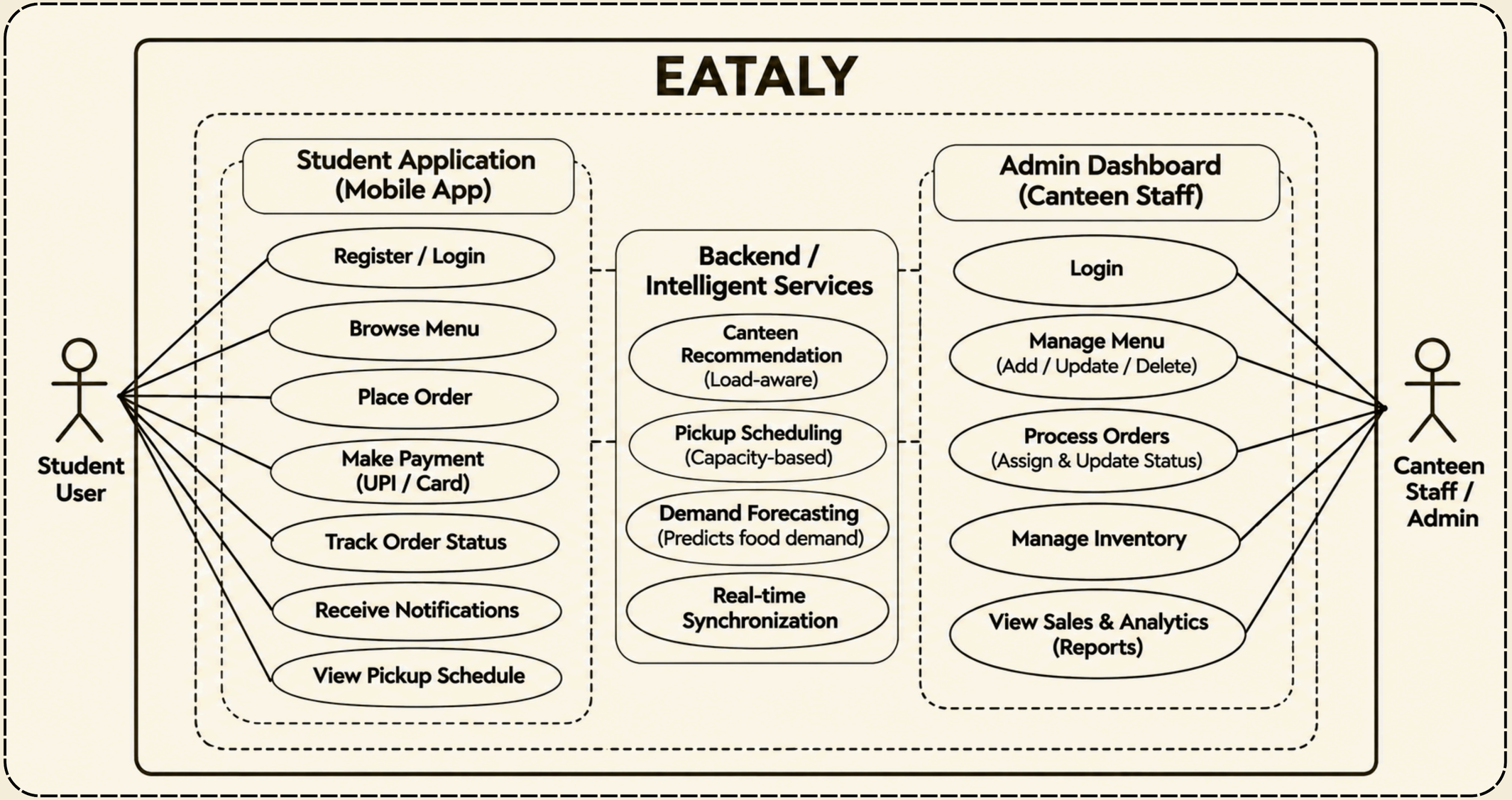


Playful character ,funky font and a bitten-cookie “L” on bold purple, showing that Eataly makes campus food fun and tasty.

APP UI DESIGN PROCESS

UI DESIGN LAYOUT → HIGH FIDELITY → FLUTTER DEVELOPMENT

ARCHITECTURE

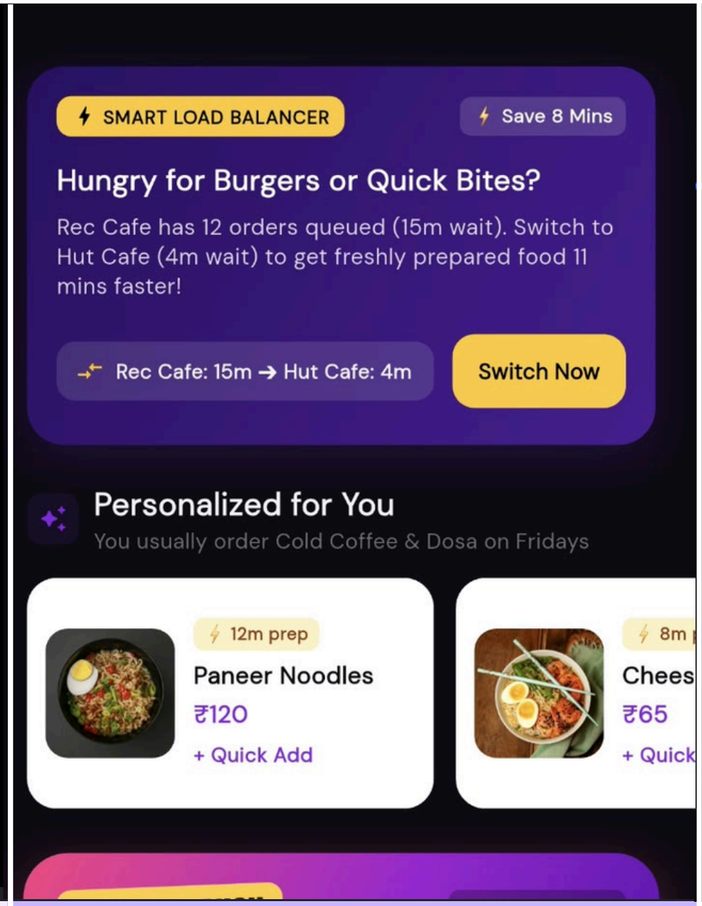


WHY EATALY ?



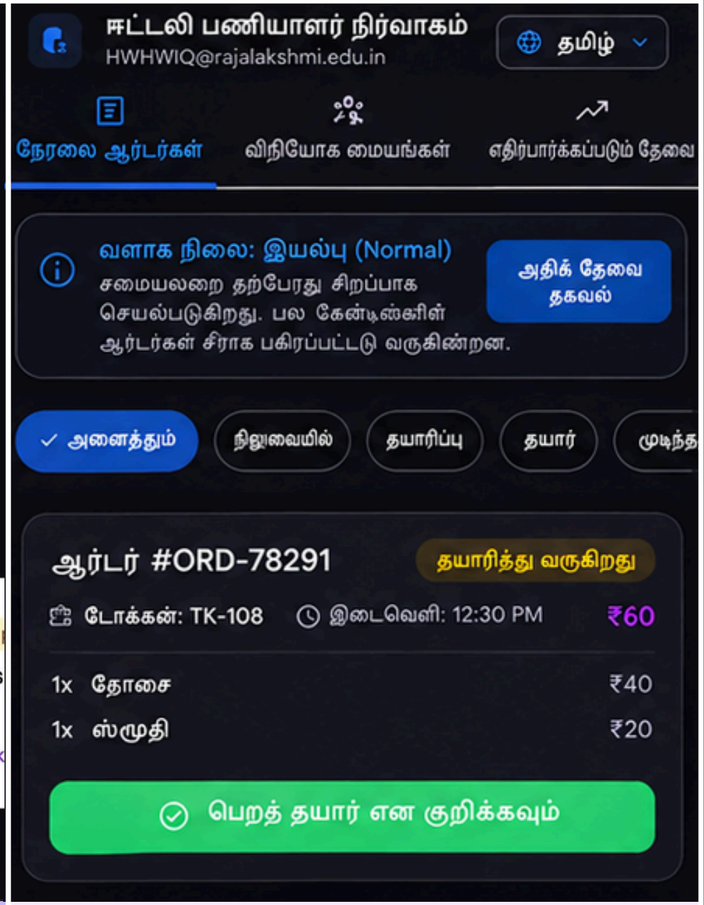
01 HEAT MAP

Visualizes food-order demand across canteen locations on the campus map. When connected to actual order data, it can help identify busy areas and improve demand planning.



02 SMART LOAD BALANCER

Distributes orders across canteens based on workload, food availability, and preparation capacity to reduce waiting time.



03 BILUNGUAL

Allows administrators to switch between English and Tamil for improved accessibility. The language preference should persist across admin pages when fully implemented.

TECHNICAL FEATURES

FIREBASE INTEGRATION

Uses Firebase services for application data and supported authentication or backend functionality.

AUTHENTICATION AND ACCESS CONTROL

Restricts administrative functions to authorized users and separates student and admin access.

REAL-TIME DATA SYNCHRONIZATION

Reflects order, menu, and stock changes across connected screens when real-time listeners are configured.

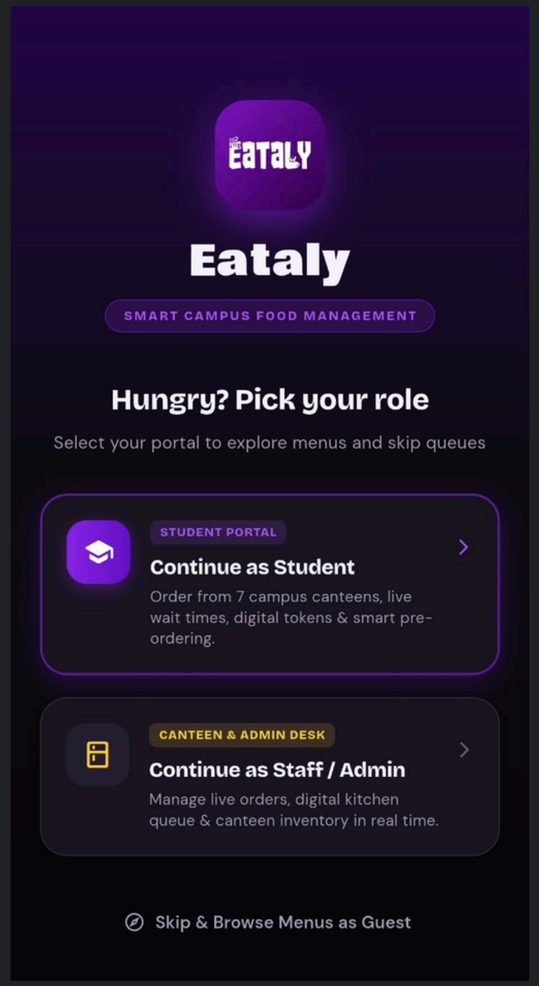
STOCK VALIDATION AT CHECKOUT

Checks item availability before confirming an order to reduce overselling and invalid orders.

FEW MORE FEATURES

ADMIN

- Sales and Performance Analytics
- Demand alerts
- Inventory and Stock Management
- Food Item Management
- Order Management



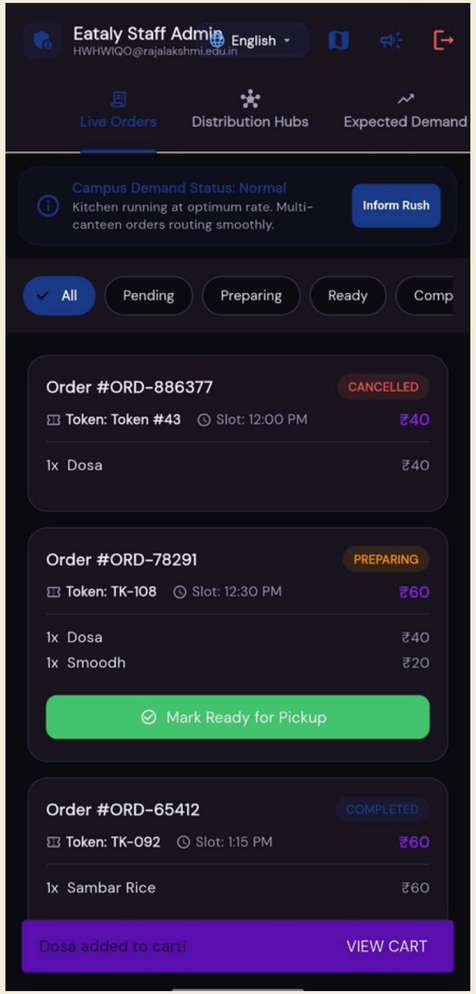
STUDENT

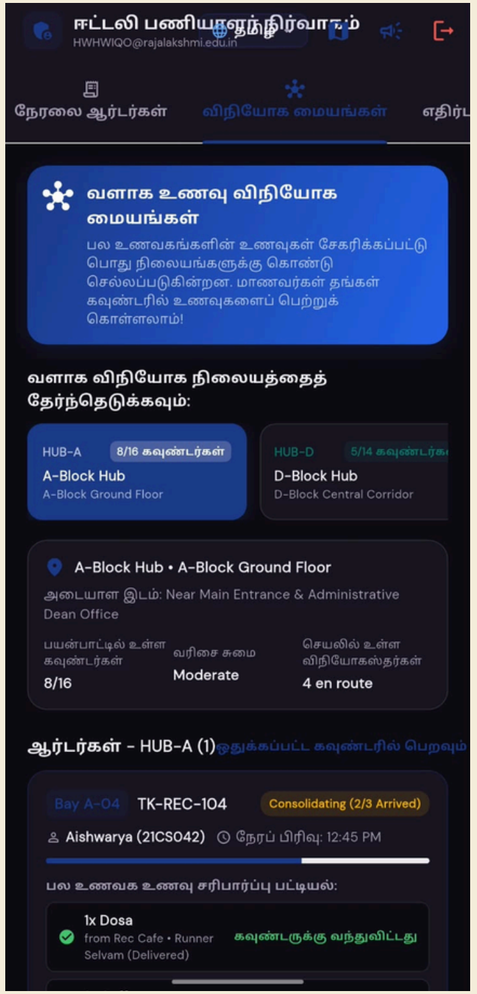
- Order Tracking and History
- Campus Map and Canteen Navigation
- Capacity-based Pickup Slots
- Digital Token
- Personalized recommendation

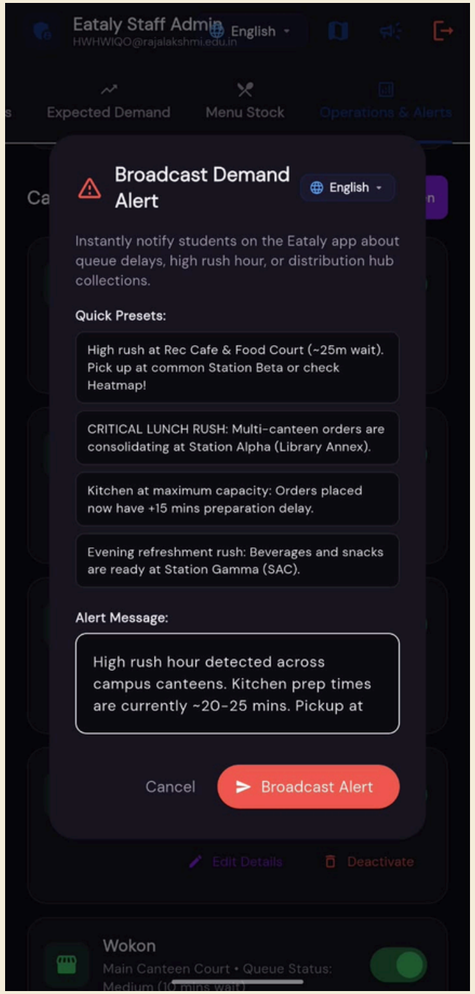
ADMIN

Features

- Secure staff login with ID and OTP verification
- Live orders by stage: Pending → Preparing → Ready → Completed
- Expected-demand view and peak-hour alerts
- Distribution hub management (runners, batches, delivery tracking)
- English/Tamil language toggle







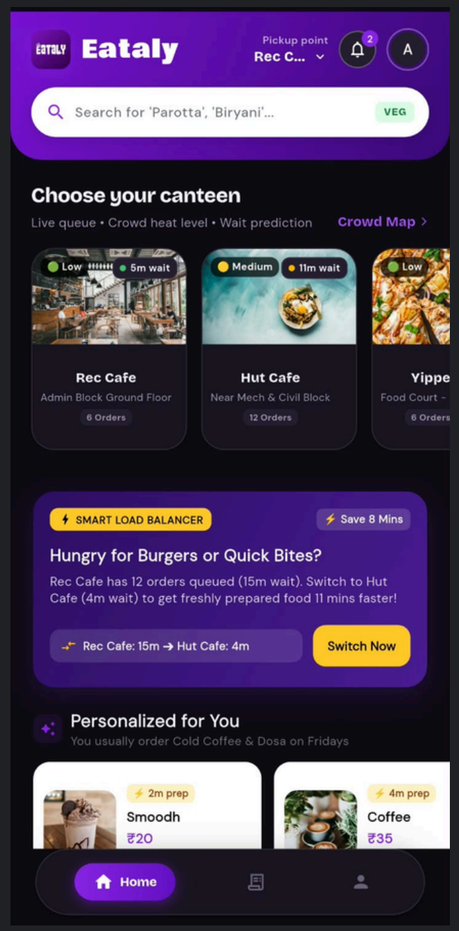
Admin verification using id | **Select foods** | **Order status update** | **Analytics**

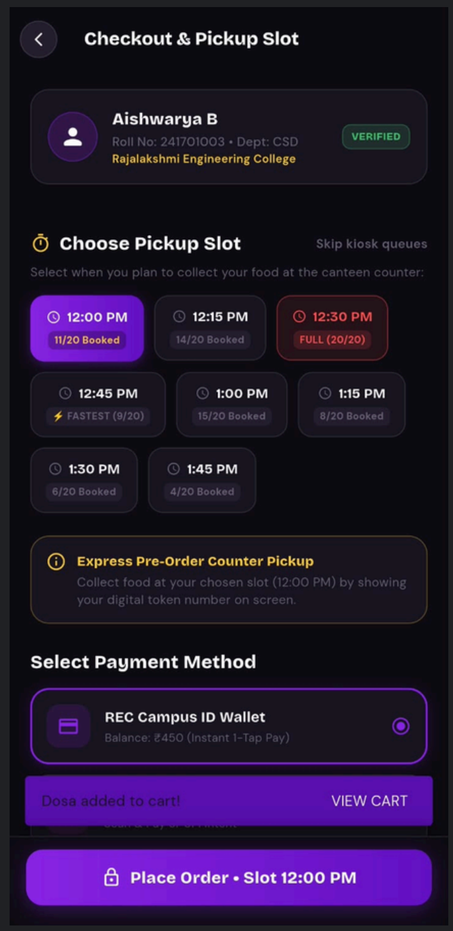
Distribution hubs | **Update and edit stock** | **Add new canteen and edit map**

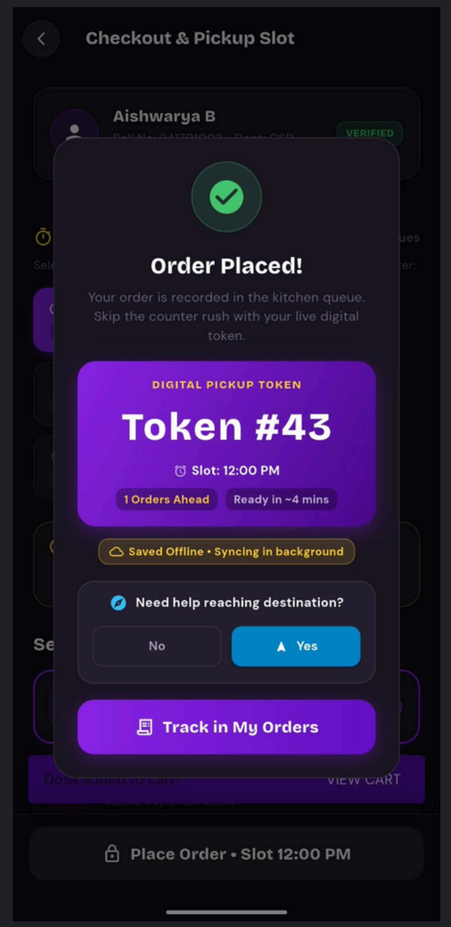
STUDENT

Features:

- Secure role-based login
- Live menu and availability
- Cart + pre-order
- Pickup slot
- token genration
- Navigation to canteen
- Order status notifications
- History + invoices







Choose canteen — **Select foods** — **Add food from other canteen** — **select pickup slot**

select payment method — **Pay** — **Token** — **Track and navigate**

THANK YOU

DO CHECK OUT OUT APP BY SCANNING THIS QR